

Group 9

CONTINUOUS DATA MEASUREMENT SYSTEM

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Problem Statement

Robyn has issues with getting continuous ice floe dynamics data in Antarctica

Factors that affect data collection:

- ▶ Infrequency of expeditions
- ▶ High-cost of expeditions
- ▶ Harsh, often dangerous environment
- ▶ Low bandwidth



Literature

- ▶ **Low-Power Wide-Area Networks: Comparison of LoRaWAN and NB-IoT Performance – IEEE Internet of Things Journal**
 - Engages in discussion about possible network types available to use
- ▶ **Internet of Extreme Things Platform for Remote Antarctic Climate Monitoring – ICMIC**
 - Features power management strategies regarding continual autonomous operation (duty cycle)
- ▶ **Ice Flow of the Antarctic Ice sheet**
 - Discusses the characteristics attributed to the area as well as the evolution of research in the region.
- ▶ **Physics of the Seasonal Ice Zone – Roach, et al**
 - Sea ice floe dynamics in the Antarctic MIZ are governed by wave-induced motion, temperature, and wind forcing — giving us exactly the physical parameters our buoy must measure.

Possible Solution: Measurement Buoy

- ▶ Measures ice position and movement & what the environment around the ice is doing
 - ▶ Acceleration, tilt, gyro
 - ▶ GPS positioning
 - ▶ Air, water, humidity sensors
- ▶ Low-power
 - ▶ Power subsystem to manage battery draw over long term deployment
- ▶ Deployable
 - ▶ Enclosure to house all the components and protect them from external env.
- ▶ Must be able to autonomously transmit data: low-bandwidth
 - ▶ Network and Comms Subsystem (LoRa) to operate in low-bandwidth env.
- ▶ Researchers must be able to access the data from the comfort of the base
 - ▶ Data is presented in GUI

